

Problem Sheet 7 Solutions

1.

(a) The integral form of the Laplace Transform is

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} f(t)e^{-st} dt,$$

$$s = \sigma + j\omega.$$

(b)

(i)

$$\mathcal{L}\{8u(t)\} = \int_0^{\infty} 8u(t)e^{-st} dt = \left[-\frac{8}{s} e^{-st} \right]_0^{\infty} = \frac{8}{s}$$

(ii)

$$\mathcal{L}\{4e^{-at}\} = \int_0^{\infty} 4e^{-at} e^{-st} dt = \int_0^{\infty} 4e^{-(s+a)t} dt = \left[\frac{-4}{s+a} e^{-(s+a)t} \right]_0^{\infty} = \frac{4}{s+a}$$

(iii) We can use the Linearity property of the Laplace transform to write

$$\begin{aligned} \mathcal{L}\{u(t) + e^{-t}u(t)\} &= \mathcal{L}\{u(t)\} + \mathcal{L}\{e^{-t}u(t)\} \\ &= \int_0^{\infty} u(t)e^{-st} dt + \int_0^{\infty} e^{-t}u(t)e^{-st} dt \\ &= \left[\frac{e^{-st}}{-s} \right]_0^{\infty} + \left[\frac{e^{-(s+1)t}}{-(s+1)} \right]_0^{\infty} \\ &= \frac{1}{s} + \frac{1}{s+1} = \frac{2s+1}{s(s+1)}. \end{aligned}$$

2.

(a) This follows directly from the linearity property of integration:

$$\begin{aligned} \mathcal{L}\{K_1 f_1(t) + K_2 f_2(t)\} &= \int_0^{\infty} (K_1 f_1(t) + K_2 f_2(t)) e^{-st} dt \\ &= \int_0^{\infty} K_1 f_1(t) e^{-st} dt + \int_0^{\infty} K_2 f_2(t) e^{-st} dt \\ &= K_1 F_1(s) + K_2 F_2(s). \end{aligned}$$

(b) To transform into the s-domain we need the LT properties for first and second derivatives (Section 15)

$$\begin{aligned} \mathcal{L}\left\{\frac{dy}{dt}\right\} &= sY - y(0^+) \\ \mathcal{L}\left\{\frac{d^2y}{dt^2}\right\} &= s^2Y - sy(0^+) - \frac{dy}{dt}(0^+). \end{aligned}$$

Since the system is stated to be 'initially relaxed' this means that $y(0^+) = 0$ and $\frac{dy}{dt}(0^+) = 0$.

Instrumentation

Taking the LT of the equation

$$s^2Y + 3sY = 2Y = 1$$
$$Y = \frac{1}{s^2 + 3s + 2} = \frac{1}{(s + 1)(s + 2)}.$$

Inverse transform (using entry 7 from the table of transforms in Section 15) gives

$$y(t) = e^{-t} - e^{-2t}.$$

(c) Similarly,

$$s^2Y - sy(0^+) - \frac{d}{dt}y(0^+) + 3sY - 3y(0^+) + 2Y = 0$$
$$Y(s^2 + 3s + 2) - y(0^+)(3 + s) - \frac{d}{dt}y(0^+) = 0.$$

Applying the known initial conditions gives

$$Y(s^2 + 3s + 2) - 5(3 + s) = 0$$
$$Y = \frac{5(3 + s)}{s^2 + 3s + 2} = \frac{5(3 + s)}{(s + 1)(s + 2)} = \frac{5(s + 1) + 10}{(s + 1)(s + 2)} = \frac{5}{s + 2} + \frac{10}{(s + 1)(s + 2)}.$$

Inverse transform yields

$$y(t) = 5e^{-2t} + 10(e^{-t} - e^{-2t})$$
$$= 10e^{-t} - 5e^{-2t}.$$

3.

(a) The transfer function of a Linear Time-Invariant (LTI) system is defined as the ratio of the Laplace transform of the output to the Laplace transform of the input. So, if the input is $y(t)$ and the output is $x(t)$ then the transfer function is

$$G(s) = \frac{\mathcal{L}\{x(t)\}}{\mathcal{L}\{y(t)\}} = \frac{X(s)}{Y(s)}.$$

A perfect voltage amplifier is a zero-order system; it has no time-delay at the output and no frequency dependence. The voltage at the output is always related to the voltage at the input by the static gain as

$$Gain = \frac{V_{out}}{V_{in}}.$$

For an amplifier with a gain of 40 dB,

$$40 = 20 \log_{10} G$$

$$Gain = 100.$$

The transfer function is therefore equal to the static sensitivity for the zero order system

$$G(s) = \frac{\mathcal{L}\{x(t)\}}{\mathcal{L}\{y(t)\}} = \frac{\mathcal{L}\{100y(t)\}}{\mathcal{L}\{y(t)\}} = 100.$$

Instrumentation

In the case that the input signal is a delta function $y(t) = \delta(t)$, then since $\mathcal{L}\{\delta(t)\} = 1$, we can state that the Transfer Function is the Laplace Transform of the system's impulse response. The output – in the time-domain – is therefore

$$\begin{aligned}x(t) &= \mathcal{L}^{-1}\{G(s)\} \\ &= 100\delta(t).\end{aligned}$$

(b) If we choose the input to the system to be the delta function $y(t) = \delta(t)$ then this has the useful property $Y(s) = 1$ which simplifies the above relation to

$$G(s) = X(s),$$

i.e. the transfer function is given by the Laplace transform of the system's impulse response. We can determine the transfer function of a system either analytically or experimentally (by providing an impulse at the input and recording the output). Once we know the transfer function then in the case of any arbitrary input $y(t)$ then the output can be determined from

$$x(t) = \mathcal{L}^{-1}\{G(s)Y(s)\}.$$

(c) Using the properties of the delta function

$$\begin{aligned}\int_{0^-}^{0^+} \delta(t) dt &= 1 \\ \delta(t) &= 0, \quad t \neq 0,\end{aligned}$$

we can see that for $t < 0$ the output is zero, and for $t > 0$ the output is 1, i.e. for $y(t) = \delta(t)$, $x(t) = u(t)$.

(d) We can state the Laplace transform of the input and output as

$$\begin{aligned}Y(s) &= \mathcal{L}\{y(t)\} = \mathcal{L}\{\delta(t)\} = 1 \\ X(s) &= \mathcal{L}\{x(t)\} = \mathcal{L}\{u(t)\} = \frac{1}{s}\end{aligned}$$

The Transfer Function is

$$G(s) = \frac{X(s)}{Y(s)} = \frac{1}{s}.$$

4.

(a) For a single amplifier

$$g_1(t) = te^{-t}.$$

The transfer function is

$$G_1(s) = \mathcal{L}\{g_1(t)\} = \mathcal{L}\{te^{-t}\} = \frac{1}{(s+1)^2}.$$

(b) For two amplifiers in series

$$G(s) = G_1(s)G_2(s) = \frac{1}{(s+1)^4}.$$

Instrumentation

(c) For a step input $y(t) = u(t)$, $Y(s) = \frac{1}{s}$,

$$x(t) = \mathcal{L}^{-1} \left\{ \frac{1}{s(s+1)^4} \right\}.$$

To find the inverse transform we need to write it as a partial fraction

$$X(s) = \frac{1}{s(s+1)^4} = \frac{k_0}{s} + \frac{k_1}{(s+1)^4} + \frac{k_2}{(s+1)^3} + \frac{k_3}{(s+1)^2} + \frac{k_4}{(s+1)}.$$

To evaluate k_0 multiply $X(s)$ by s and evaluate for $s = 0$

$$\left. \frac{1}{(s+1)^4} \right|_{s=0} = k_0 = 1.$$

To evaluate k_1 multiply $X(s)$ by $(s+1)^4$ and evaluate for $s = -1$

$$\left. \frac{1}{s} \right|_{s=-1} = k_1 = -1.$$

For k_2, k_3, k_4 we cannot use $s = -1$ again directly as we have already used this to find k_1 . However, since the partial fraction is an *identity* we can differentiate it. First, we can simplify the partial fraction and apply the known constants k_0 and k_1

$$(s+1)^4 X(s) = \frac{1}{s} = \frac{(s+1)^4}{s} - 1 + k_2(s+1) + k_3(s+1)^2 + k_4(s+1)^3.$$

Differentiating yields

$$-\frac{1}{s^2} = 3s^2 + 8s + 6 - \frac{1}{s^2} + k_2 + k_3(2s+2) + k_4(3s^2 + 6s + 3).$$

Now we can use the $s=-1$ to get k_2

$$k_2 = -1$$

Differentiating again yields k_3

$$k_3 = -1$$

And again for k_4

$$k_4 = -1$$

Therefore

$$\begin{aligned} x(t) &= \mathcal{L}^{-1} \left\{ \frac{1}{s} - \frac{1}{s+1} - \frac{1}{(s+1)^2} - \frac{1}{(s+1)^3} - \frac{1}{(s+1)^4} \right\} \\ &= u(t) - e^{-t} - te^{-t} - \frac{t^2 e^{-t}}{2!} - \frac{t^3 e^{-t}}{3!}. \end{aligned}$$